

symbolicFK.m Documentation

symbolicFK.m

Purpose

Derives the robot Forward Kinematics **symbolically**:

$$(q_1, q_2, \dots, q_6) \rightarrow {}^B T_{TCP}(q)$$

Unlike `controlFK.m`, which calculates FK for numerical joint angles, this file produces the general FK equations.

Used for:

- Inspecting the FK equations
- Mathematical verification
- Supporting Jacobian / IK derivations
- Cross-checking `controlFK.m`

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Main Process

1. Define Symbolic Joint Angles

```
syms q1 q2 q3 q4 q5 q6 real
```

with:

$$q = [q_1 \quad q_2 \quad q_3 \quad q_4 \quad q_5 \quad q_6]^T$$

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2. Raw Symbolic FK

The normal robot DH parameters are used to build:

$$T_{raw} = A_1 A_2 A_3 A_4 A_5 A_6^F T_{TCP}$$

`T_raw` is kept as the original symbolic reference.

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3. Clean Symbolic FK

A second FK chain is created using:

- Rational symbolic link dimensions
- Exact symbolic twist angles such as $\pi/2$

This avoids floating-point artifacts in the symbolic expressions.

```
a_clean = sym(a, 'r');  
d_clean = sym(d, 'r');
```

The result is:

```
T_clean
```

with:

$${}^B T_{TCP} = \begin{bmatrix} R_{B,TCP} & p_{B,TCP} \\ 0 & 1 \end{bmatrix}$$

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Final Symbolic TCP Position

Define:

$$q_{23} = q_2 + q_3$$
$$q_{234} = q_2 + q_3 + q_4$$

Then:

$$p_{TCP} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

where:

X Position

$$\begin{aligned} x = & \frac{2183}{20000} \sin q_1 - \frac{17}{40} \cos q_1 \cos q_2 + \frac{823}{10000} \cos q_5 \sin q_1 \\ & - \frac{823}{10000} \cos q_{234} \cos q_1 \sin q_5 \\ & + \frac{1893}{20000} \cos q_1 \sin q_{234} \\ & - \frac{1569}{4000} \cos q_1 \cos q_{23} \end{aligned}$$

Y Position

$$\begin{aligned} y = & -\frac{823}{10000} \cos q_1 \cos q_5 - \frac{17}{40} \cos q_2 \sin q_1 - \frac{2183}{20000} \cos q_1 \\ & - \frac{823}{10000} \cos q_{234} \sin q_1 \sin q_5 \\ & + \frac{1893}{20000} \sin q_1 \sin q_{234} \\ & - \frac{1569}{4000} \sin q_1 \cos q_{23} \end{aligned}$$

Z Position

$$\begin{aligned} z = & 0.089159 - \frac{17}{40} \sin q_2 - \frac{1569}{4000} \sin q_{23} \\ & - \frac{1893}{20000} \cos q_{234} - \frac{823}{10000} \sin q_5 \sin q_{234} \end{aligned}$$

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Final Symbolic TCP Orientation

Let:

$$\phi = q_2 + q_3 + q_4$$

Then:

$$R_{B,TCP} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

with:

$$\begin{aligned} r_{11} &= \cos q_6 (\sin q_1 \sin q_5 + \cos \phi \cos q_1 \cos q_5) - \sin \phi \cos q_1 \sin q_6 \\ r_{12} &= -\sin q_6 (\sin q_1 \sin q_5 + \cos \phi \cos q_1 \cos q_5) - \sin \phi \cos q_1 \cos q_6 \\ r_{13} &= \cos q_5 \sin q_1 - \cos \phi \cos q_1 \sin q_5 \\ r_{21} &= -\cos q_6 (\cos q_1 \sin q_5 - \cos \phi \cos q_5 \sin q_1) - \sin \phi \sin q_1 \sin q_6 \\ r_{22} &= \sin q_6 (\cos q_1 \sin q_5 - \cos \phi \cos q_5 \sin q_1) - \sin \phi \cos q_6 \sin q_1 \\ r_{23} &= -\cos q_1 \cos q_5 - \cos \phi \sin q_1 \sin q_5 \\ r_{31} &= \cos \phi \sin q_6 + \sin \phi \cos q_5 \cos q_6 \\ r_{32} &= \cos \phi \cos q_6 - \sin \phi \cos q_5 \sin q_6 \\ r_{33} &= -\sin \phi \sin q_5 \end{aligned}$$

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Validation

The clean symbolic FK is evaluated at:

```
q_test = [
    0.2
   -0.3
    0.5
    0.1
   -0.4
    0.2
];
```

and compared against `T_raw`.

Maximum difference:

$$5.5511 \times 10^{-17}$$

which is floating-point round-off.

Therefore:

$$T_{clean} \approx T_{raw}$$

The symbolic FK has also been cross-validated with `controlFK.m`.

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Related Files

- `dhTransform.m` — Standard DH transformation
- `controlFK.m` — Numerical runtime FK
- `controlJacobian.m` — Numerical geometric Jacobian
- `symbolicJacobian.m` — Symbolic Jacobian